Compact MMI Couplers and a Double-Racetrack Filter on 220-nm SOI

openEBL 2026-10 design report – layout EBeam_25DR0041.oas

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1. Introduction

Multimode-interference (MMI) couplers are the workhorse power splitters and combiners of silicon photonics. Based on the self-imaging principle [1, 2], they provide broadband, polarization-robust splitting with relaxed fabrication tolerances compared with directional couplers, and they are the building blocks of Mach–Zehnder switches and modulators [4], coherent receivers and the splitter trees of optical phased arrays [9]. Compact MMI libraries on SOI [3] and footprint-reduced splitters [5] show that the size can be reduced to a few tens of micrometres, but the split ratio, excess loss and bandwidth of short MMIs become increasingly sensitive to the etch bias and the silicon thickness.

This chip addresses that trade-off with two approaches. First, a robust 2×2 MMI whose sidewall was shape-optimized by adjoint inverse design [6] with a multi-fidelity (2.5D/3D) model and several fabrication corners in the figure of merit. Second, three compact MMIs (two 2×2, one 3×3) designed with Lumerical simulations. A double-racetrack add-drop filter [7] completes the chip as a wavelength-selective element and provides an on-chip reference for the grating-coupler response. All circuits follow the SiEPIC design-for-test rules [8].

2. Overview

The layout contains five independent test circuits on the 605 × 410 µm openEBL die (SiEPIC EBeam PDK, 220-nm Si, full etch, oxide cladding, TE 1550 nm). Four are multimode-interference (MMI) couplers – one adjoint-optimized robust 2×2 MMI and three designs from Lumerical models (two 2×2, one 3×3) – and the fifth is a double-racetrack add-drop filter. Every device is measured as a single component (no MZI), so the measured spectra give its transmission, split ratio and bandwidth directly.

#	Device	Footprint (body)	opt_in label (TE, 1550 nm)	Measured ports
1	Robust adjoint 2×2 MMI (robust_c)	10.5 × 3.0 µm	opt_in_TE_1550_device_25DR0041_MMI_robust_c	bar, cross
2	2×2 MMI D01	26.3 × 2.7 µm	opt_in_TE_1550_device_25DR0041_MMI_D01	bar, cross
3	Double racetrack add-drop	2 rings, R 10.8 µm	opt_in_TE_1550_device_25DR0041_DoubleRacetrack_R10p8_Lc2_g100nm	through, drop
4	3×3 MMI D02	56 × 4.8 µm	opt_in_TE_1550_device_25DR0041_MMI3x3_D02	3 outputs*
5	2×2 MMI D03	41.6 × 3.4 µm	opt_in_TE_1550_device_25DR0041_MMI_D03	bar, cross

** all three outputs when the four-GC column fits in the die height, otherwise the two outer outputs (the layout script decides and reports it).*

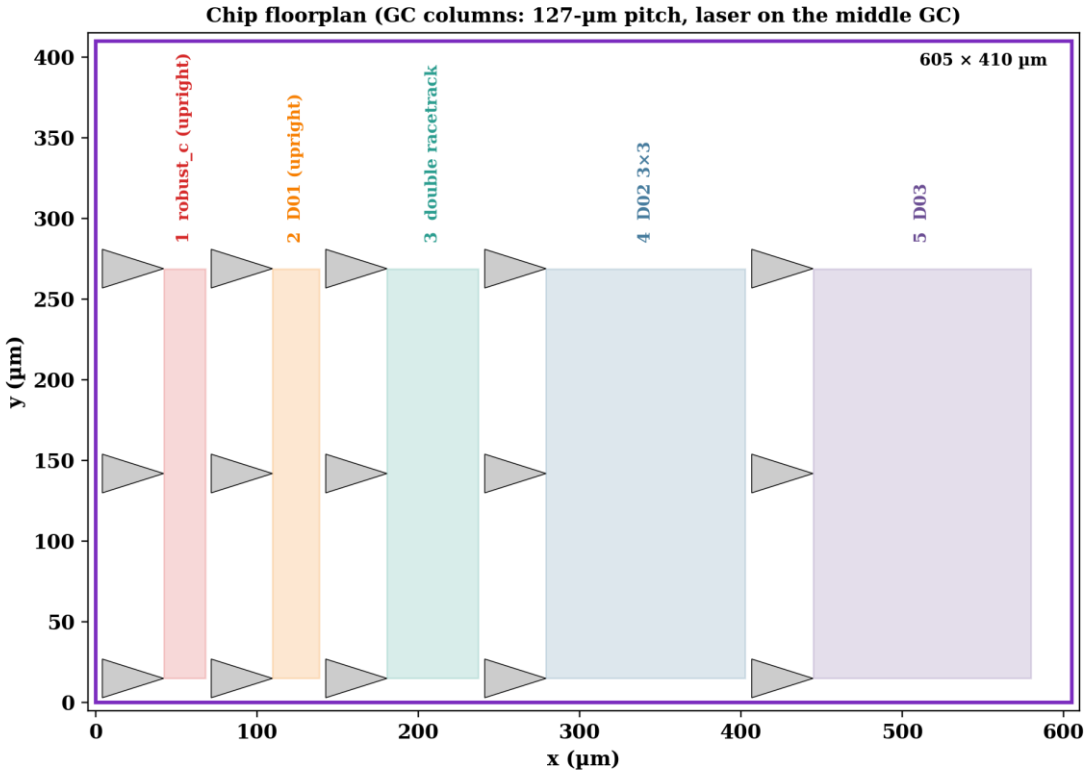


Figure 1. Chip floorplan (schematic). Devices 1 and 2 stand upright between the middle and the top grating coupler to save die width.

3. Device 1 – robust adjoint-optimized 2×2 MMI

A 3.0 × 10.5 µm paired-interference MMI whose sidewall (12-point PCHIP spline, mirror-symmetric) was shape-optimized with a multi-fidelity adjoint method: 2.5D varFDTD gradients corrected by 3D FDTD output space mapping, with five fabrication corners (etch bias ±20 nm, Si 210/230 nm) in one robust figure of merit. Tapers 5 µm (0.5 → 0.867 µm), taper gap 150 nm at the facet, 5-µm cosine S-bends to a 2-µm port pitch.

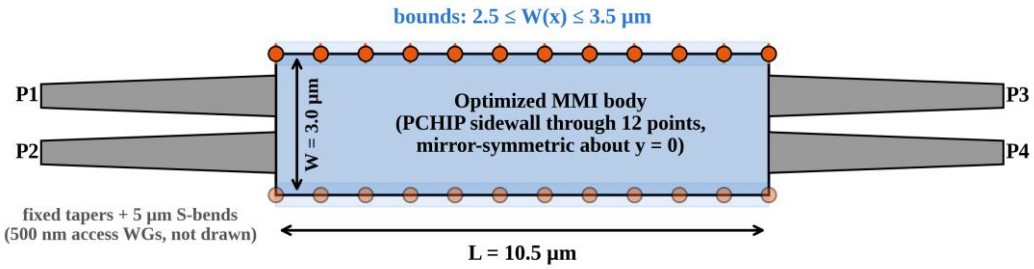


Figure 2. Parametrization of the robust MMI (12 sidewall control points, fixed tapers and S-bends).

Final 3D verification (Lumerical FDTD, Palik Si and SiO_2 with a tight dispersion fit, mesh accuracy 4, 15-nm override in x, y and z; C-band 1530–1565 nm):

Corner	IL @1550 (dB)	Imbalance @1550 (dB)	C-band worst IL (dB)	C-band worst |imb| (dB)	Phase err (°)	0.3-dB BW (nm)
bias −20 nm	0.174	+0.114	0.286	0.187	1.7	67
nominal	0.085	+0.091	0.160	0.136	0.8	83
bias +20 nm	0.036	+0.052	0.081	0.139	0.3	76
Si 210 nm	0.165	−0.086	0.259	0.174	1.9	67
Si 230 nm	0.059	+0.137	0.092	0.195	0.6	92

Nominal split at 1550 nm: 50.5 : 49.5. All corners meet IL < 0.3 dB, |imbalance| < 0.3 dB and phase error < 5°.

4. Device 2 – 2×2 MMI D01

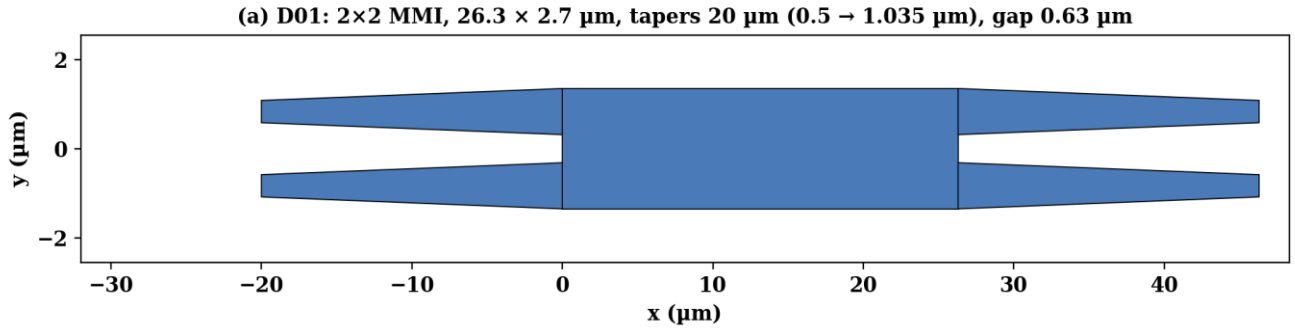


Figure 3. Drawn geometry of the three Lumerical-model MMIs: (a) D01, (b) D03, (c) D02 (3×3).

Parameter	Value
MMI body (L × W)	26.3 × 2.7 µm
Tapers	linear, 20 µm, 0.5 → 1.035 µm
Port offset / gap between tapers	±0.8325 µm / 0.63 µm
Excess loss (simulated)	0.07 ± 0.01 dB around 1550 nm (minimum 0.07 dB at 1550 nm; 0.41 dB at 1500 nm, 0.28 dB at 1600 nm)
Output transmissions S21 / S31	0.44–0.495 each over 1500–1600 nm; balanced (≈ 0.49 / 0.49) at 1540–1560 nm
Wavelength range	1500–1600 nm

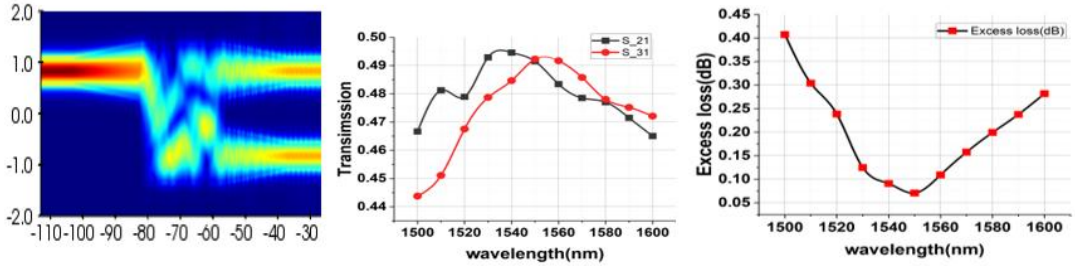


Figure 4. Simulated D01 2×2 MMI splitter: optical intensity, output transmissions S21/S31 and excess loss vs wavelength.

5. Device 3 – double-racetrack add-drop filter

Two coupled racetracks between two bus waveguides (port 1 input, port 2 through, port 3 drop, port 4 add – terminated). Bend radius 10.8 µm (waveguide centre line), straight coupling sections 2 µm, all three gaps (bus–ring, ring–ring, ring–bus) 100 nm, w = 500 nm. Round trip 71.9 µm, expected FSR ≈ 7.9 nm ($n_g \approx 4.2$), in agreement with the simulated resonance spacing of ≈ 8 nm (Figure 6). At resonance the simulated drop transmission reaches ≈ 0.95 with a through-port dip to ≈ 0.08; off resonance the drop port stays 40–50 dB below. Off resonance, the through-port transmission also calibrates the grating-coupler loss (the die has no room for a separate loopback).

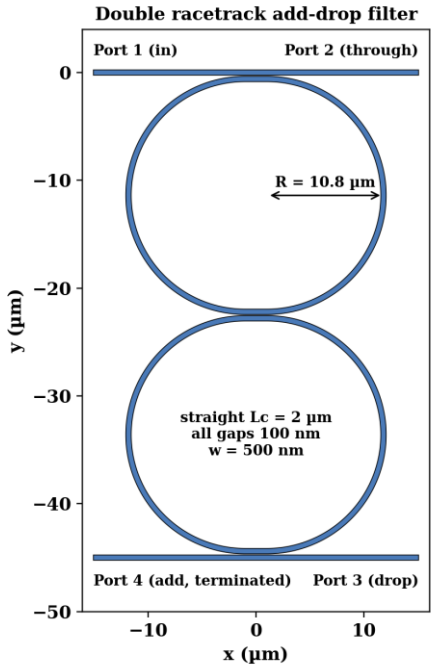


Figure 5. Double-racetrack geometry.

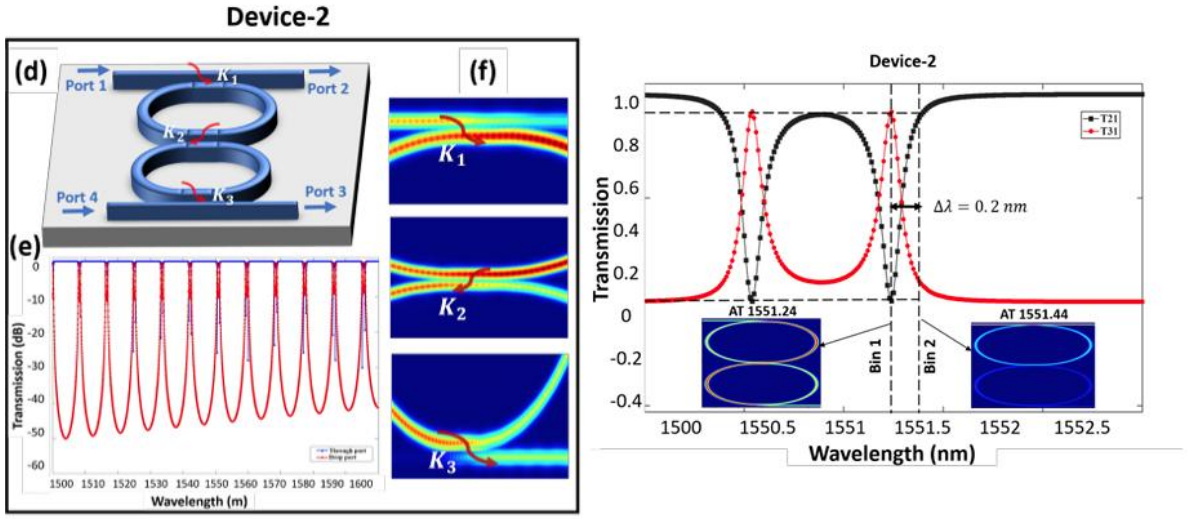
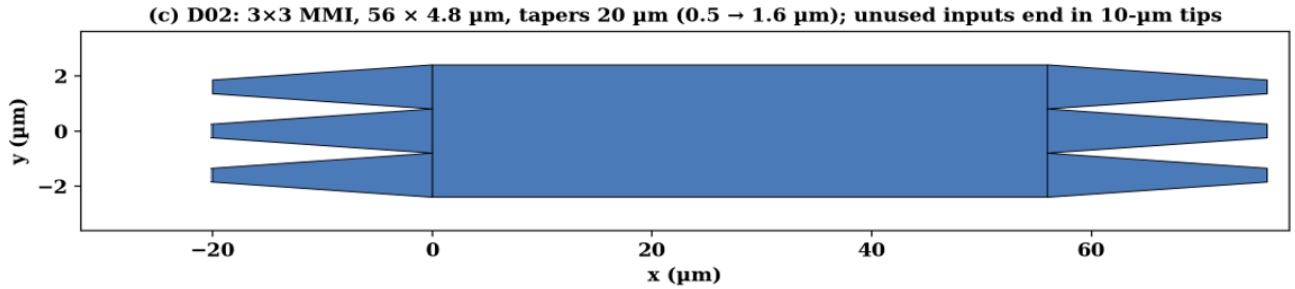


Figure 6. Simulated double racetrack: through (T21) and drop (T31) spectra over 1500–1600 nm with the coupling regions K1–K3, and a zoom near 1551 nm (drop peak ≈ 0.95, through dip ≈ 0.08, 0.2-nm spacing between bins).

6. Device 4 – 3×3 MMI D02



Parameter	Value
MMI body (L × W)	56 × 4.8 µm
Tapers	linear, 20 µm, 0.5 → 1.6 µm, centred at 0, ±1.6 µm
Input / unused inputs	top-left input; the other two inputs end in 10-µm inverse tapers (80-nm tip)

Note	3 × 1.6 µm = W: the tapers touch at the facet (gap 0)
Excess loss (simulated)	0.086 ± 0.005 dB (minimum 0.035 dB at 1530 nm; 0.15 dB at 1500 nm, 0.33 dB at 1590 nm)
Output transmissions S21 / S31 / S41	≈ 0.33 / 0.32 / 0.33 at 1545 nm; S21 flat (0.33), S31 0.28 → 0.33, S41 0.355 → 0.25 over 1500–1600 nm
Wavelength range	1500–1600 nm

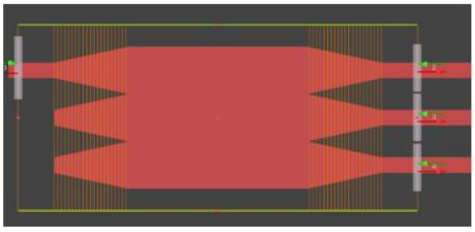


Fig.5 Simulated structure of 3x3 MMI splitter

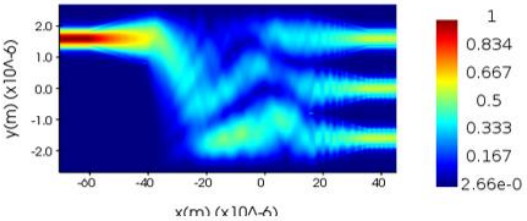


Fig.6 Optical intensity of 3x3 MMI splitter

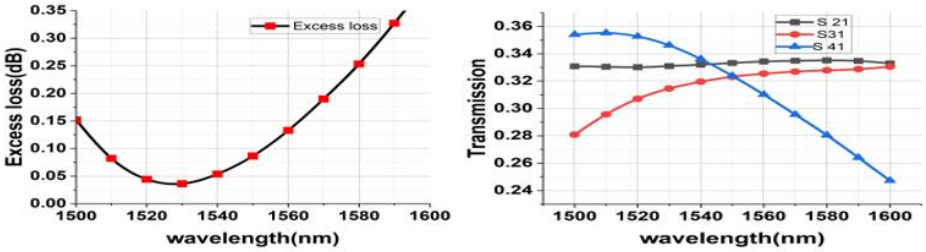
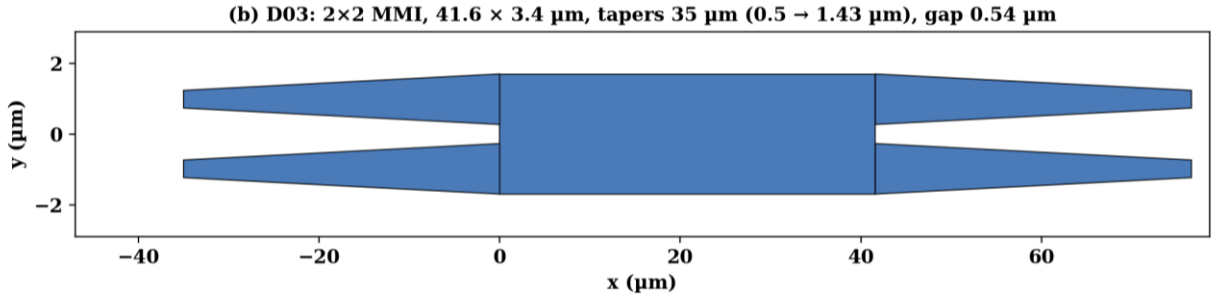


Figure 7. Simulated D02 3×3 MMI splitter: structure, optical intensity, excess loss and output transmissions.

7. Device 5 – 2×2 MMI D03



Parameter	Value
MMI body (L × W)	41.6 × 3.4 µm
Tapers	linear, 35 µm, 0.5 → 1.43 µm
Port offset / gap between tapers	±0.985 µm (tapers flush with the side walls) / 0.54 µm
Excess loss (simulated)	minimum 0.056 dB at 1540 nm; ≤ 0.10 dB over ≈ 1520–1570 nm; 0.24 dB at 1500 nm, 0.19 dB at 1600 nm
Normalized output power (upper / lower arm)	0.465–0.499 each over 1500–1600 nm; ≈ 0.492 / 0.491 at 1550 nm (balanced within ≈ 0.01)
Wavelength range	1500–1600 nm

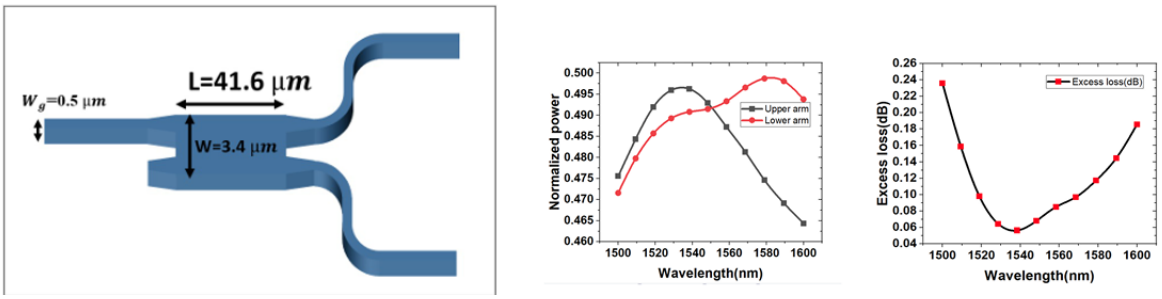


Figure 8. D03 2×2 MMI power splitter: schematic, normalized output power of the two arms and excess loss vs wavelength.

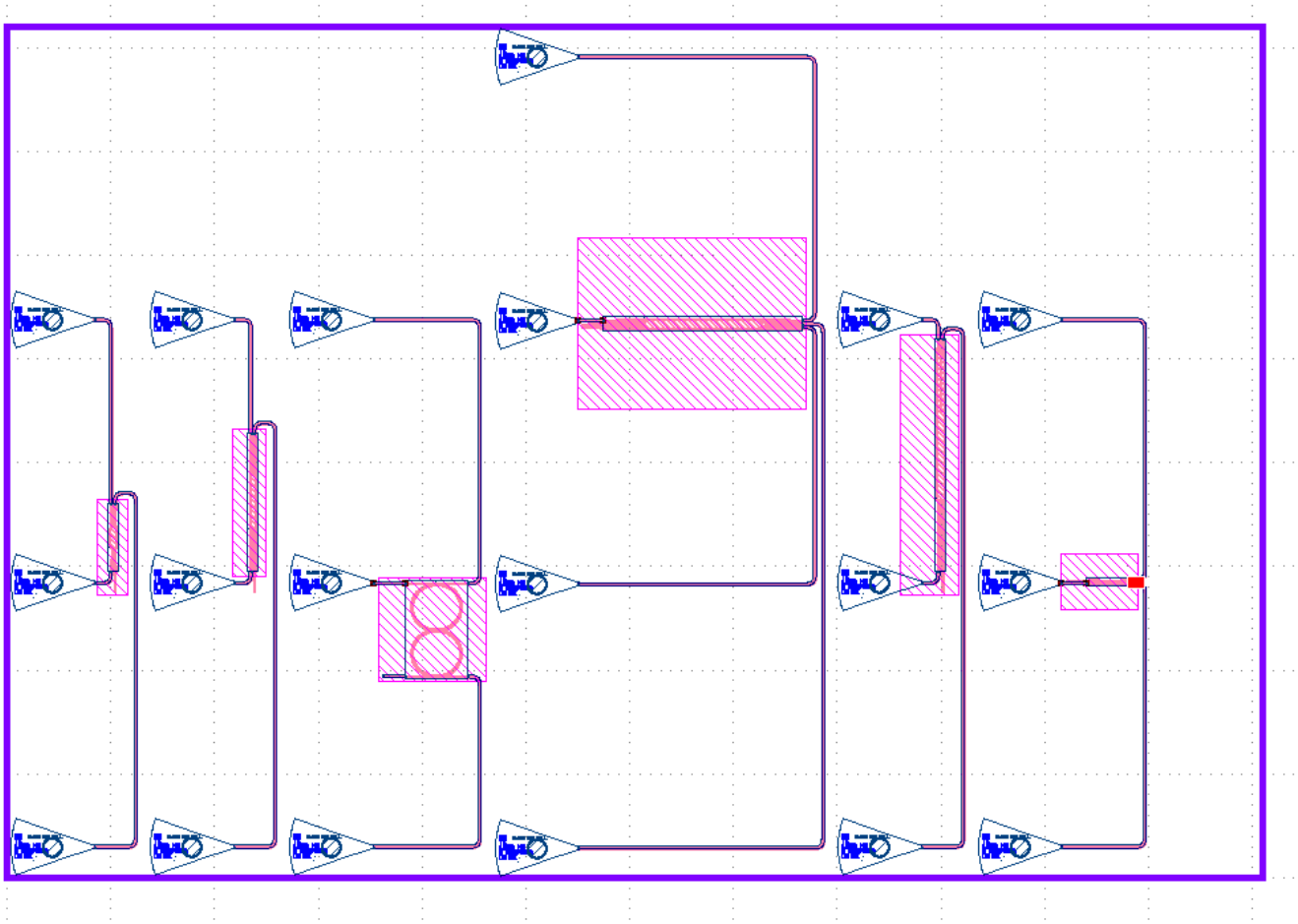
8. Simulated performance summary

Metric	1: robust_c 2×2	2: D01 2×2	3: double racetrack	4: D02 3×3	5: D03 2×2
Body size (μm)	10.5 × 3.0	26.3 × 2.7	R 10.8, Lc 2	56 × 4.8	41.6 × 3.4
Loss	IL 0.085 dB @1550 (C-band worst 0.16 dB)	excess 0.07 ± 0.01 dB	drop ≈ 0.95 at resonance	excess 0.086 ± 0.005 dB	excess 0.056 dB @1540 (≤ 0.1 dB, 1520–1570)
Split / response	50.5 : 49.5 @1550	≈ 50 : 50 @1540–1560	FSR ≈ 8 nm	≈ 1/3 each @1545	≈ 50 : 50 @1550
Fab tolerance (3D)	5 corners, all pass	–	–	–	–
Range	1500–1600 nm	1500–1600 nm	1500–1600 nm	1500–1600 nm	1500–1600 nm

9. Layout and design-for-test

- Grating couplers GC_TE_1550_8degOxide_BB in vertical columns on a 127-μm pitch, all facing +x; laser on the middle GC, one detector above and one (two for the 3×3) below; opt_in labels at the input GCs; circuits ≥ 61 μm apart.
- Waveguides: Strip TE 1550 nm, w = 500 nm, bend radius 5 μm. Unused MMI inputs end in built-in 10-μm inverse tapers (no reflection, no extra footprint).
- Space saving: devices 1 and 2 are rotated by 90° and placed between the middle and the top GC, which reduces each circuit width from ~100 μm to ~28 μm; all five devices fit within x = 0–580 μm.
- SEM request boxes (layer 200/0, 4:3 landscape) each cover a complete device – tapers, MMI body, port waveguides and termination tips, and both racetracks with both buses – with a 2-μm margin (robust_c 62 × 46.5 μm, D01 107 × 80 μm, ring 66 × 49.5 μm, D02 110 × 82.5 μm, D03 116 × 87 μm), to compare the fabricated and drawn shapes.
- Top cell and file name: EBeam_25DR0041. SiEPIC functional verification: 0 errors (confirm after the final KLayout run).

[final KLayout screenshot of the full layout here]



10. Test plan

- Laser sweep 1500–1600 nm (TE), one sweep per circuit; detectors on all output GCs of the circuit.
- MMIs: split ratio T_cross/(T_bar+T_cross) and imbalance vs wavelength (independent of the GC response, which is common to both outputs); excess loss after subtracting the GC response obtained from the ring through port off resonance.
- 3×3 MMI: power uniformity among the outputs and total transmission.
- Double racetrack: resonance wavelengths, FSR, extinction and box-like drop response; group index from FSR; coupling from the drop-port line shape.
- Device 1 is compared with its corner simulations (Section 3) to validate the robust multi-fidelity adjoint design.

References

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